

Which Quality Tool Fits Best?

10 Real-World Scenarios • Identify the Right Tool • Solutions Included

Process Map

Check Sheet

Histogram

Scatter Plot

Control Chart

Fishbone

Pareto

Affinity

Tree Diagram

PERT

How to Use This Deck

1

Read the Scenario

Each question slide describes a real business problem without naming the tool.

2

Discuss & Decide

As a class, decide which quality tool from B7 or N7 is the best fit — and why.

3

Check the Solution

The answer slide follows immediately after each question, with a full justification.

4

Reflect

Consider: could any other tool also apply? What makes the chosen tool the strongest choice?

Emergency Room Overcrowding

SCENARIO

A large urban hospital in Riyadh is struggling with long ER wait times. The hospital administrator suspects there are multiple bottlenecks — from patient check-in, through triage, physician assessment, lab tests, and bed assignment — but nobody has ever documented the entire sequence of steps end-to-end.

? The administrator wants to see the full sequence of activities, decision points, and handoffs between departments before attempting any improvement. What quality tool should be used first?

OPTIONS

A) Pareto Chart

B) Control Chart

C) Process Map

D) Fishbone Diagram

Emergency Room Overcrowding



CORRECT ANSWER: C) Process Map

WHY THIS TOOL?

A Process Map (flowchart) is the ideal first step when a complex multi-step process has never been documented. It reveals all activities, decision points, and hand-offs visually — showing exactly where delays occur before any data collection begins. A fishbone assumes you already know the problem; a Pareto assumes you have frequency data; a control chart monitors an existing stable process.



REAL-WORLD OUTCOME

After mapping, the team discovered 4 unnecessary sign-off steps that added 47 minutes to average wait times.

Ceramic Tile Defects at a Factory

SCENARIO

A ceramic tile manufacturer in Jeddah is receiving customer complaints about defective tiles. The quality manager suspects several defect types are occurring — chipped edges, colour inconsistency, warping, size deviation, and surface cracks — but doesn't know which defect is most common. Over two weeks, inspectors manually check every 50th tile off the production line.

? Inspectors need a simple, structured way to record each defect type every time they encounter it during their shift. Which tool is most appropriate for this data collection stage?

OPTIONS

A) Histogram

B) Check Sheet

C) Control Chart

D) Scatter Diagram

Ceramic Tile Defects at a Factory



CORRECT ANSWER: B) Check Sheet

WHY THIS TOOL?

A Check Sheet is specifically designed for structured, real-time defect counting by category. Inspectors simply place a tick mark in the corresponding defect row each time they spot it. This produces clean frequency data by defect type — which can then directly feed a Pareto Chart. A histogram requires already-collected continuous data; a control chart monitors process stability over time.



REAL-WORLD OUTCOME

The check sheet data revealed chipped edges accounted for 61% of all defects — a finding invisible without structured recording.

Coffee Cup Fill Weights

SCENARIO

A global coffee chain's regional bottling plant fills 250ml take-away cups by machine. The operations team collects 200 fill weight measurements (in ml) from one production shift. They want to understand whether fills are centred near 250ml, how much variation exists, and whether the distribution is symmetric — before deciding if the machine needs recalibration.

? The team has 200 continuous numerical measurements and wants to visualise the shape, spread, and centre of the fill weight distribution. Which tool directly serves this need?

OPTIONS

A) Pareto Chart

B) Scatter Diagram

C) Histogram

D) Affinity Diagram

Coffee Cup Fill Weights



CORRECT ANSWER: C) Histogram

WHY THIS TOOL?

A Histogram is built precisely for visualising the distribution of continuous numerical data. It shows whether fills are centred at 250ml, reveals skewness or bimodal patterns, and highlights outliers — all in a single visual. A Pareto chart requires categorical defect data; a scatter diagram requires two paired variables; an affinity diagram is for organising qualitative ideas.



REAL-WORLD OUTCOME

The histogram exposed a left skew — indicating the machine was systematically under-filling, violating consumer protection regulations.

Overtime & Absenteeism at a Fulfilment Centre

SCENARIO

A large e-commerce fulfilment centre in Dubai notices that some warehouse departments have much higher absenteeism rates than others. HR suspects that mandatory overtime hours may be driving staff burnout and absence. They pull data for 60 employees: total overtime hours worked last month and number of days absent that month.

? HR wants to visually determine whether a statistical relationship exists between overtime hours worked and employee absenteeism. Which tool is most appropriate?

OPTIONS

A) Control Chart

B) Histogram

C) Check Sheet

D) Scatter Diagram

Overtime & Absenteeism at a Fulfilment Centre



CORRECT ANSWER: D) Scatter Diagram

WHY THIS TOOL?

A Scatter Diagram plots two continuous variables against each other to reveal whether a relationship (correlation) exists. By plotting overtime hours (x-axis) vs. days absent (y-axis) for each employee, the team can immediately see if the points trend upward (positive correlation). The other tools deal with single-variable distributions or process monitoring — none are designed to show the relationship between two variables.



REAL-WORLD OUTCOME

The scatter plot revealed a strong positive correlation ($r = 0.78$), leading management to cap overtime at 15hrs/week — absenteeism dropped 34% in 3 months.

Tablet Coating Thickness Drift

SCENARIO

A pharmaceutical company coats tablets with an active-ingredient layer that must be between 0.18mm and 0.22mm thick. Every hour, a technician samples 5 tablets and records average thickness. Over 30 days, management notices occasional customer complaints about coating being too thin, but they cannot tell if the process is drifting or if the variation is random noise.

? Management needs to distinguish between random (common cause) variation and assignable (special cause) variation in the coating process over time. Which tool is best suited?

OPTIONS

A) Pareto Chart

B) Fishbone Diagram

C) Control Chart

D) Tree Diagram

Tablet Coating Thickness Drift



CORRECT ANSWER: C) Control Chart

WHY THIS TOOL?

A Control Chart is the only tool designed specifically to monitor process stability over time by plotting measurements against statistically calculated Upper and Lower Control Limits (UCL/LCL). Points outside the limits signal special causes that need investigation. The other tools do not provide this time-series statistical monitoring capability — they are used for identifying defect causes or planning, not process stability monitoring.



REAL-WORLD OUTCOME

The X-bar chart revealed a systematic downward drift every afternoon — traced to a coating drum that overheated when ambient temperature exceeded 32°C.

Flight Departure Delays

SCENARIO

A regional airline in the Gulf is experiencing high rates of departure delays. The operations team has collected 6 months of delay records, categorised by root cause: late crew, fuelling delay, baggage handling, catering delay, technical fault, late passengers, gate conflict, and weather. A total of 840 delay events were recorded across these 8 categories.

? The airline wants to identify which few delay causes are responsible for the majority of total delays, so they can focus their improvement efforts most efficiently. Which tool is best?

OPTIONS

A) Pareto Chart

B) Scatter Diagram

C) Process Map

D) Affinity Diagram

Flight Departure Delays



CORRECT ANSWER: A) Pareto Chart

WHY THIS TOOL?

A Pareto Chart applies the 80/20 principle — it ranks categorical defect/delay causes by frequency (descending bars) and overlays a cumulative percentage line. This directly identifies the 'vital few' causes driving 80% of delays. Scatter diagrams require two numeric variables; process maps document workflows; affinity diagrams organise brainstormed ideas. Only the Pareto chart is built to prioritise categorical frequency data.



REAL-WORLD OUTCOME

The Pareto revealed that just 2 causes — baggage handling (38%) and late crew (29%) — caused 67% of all delays, focusing corrective action precisely.

Concrete Strength Failures

SCENARIO

A construction company in Mecca is experiencing recurring concrete compressive strength failures on a major project. The site engineer calls a team meeting with the structural engineer, QC inspector, lab technician, procurement officer, and site supervisor. They know the problem but are struggling to agree on what is truly causing it — discussions keep cycling back to symptoms rather than root causes.

? The team needs a structured brainstorming tool that systematically organises potential causes into major categories (Man, Machine, Material, Method, Environment) to find the root cause. Which tool fits?

OPTIONS

A) Control Chart

B) Check Sheet

C) Fishbone (Ishikawa) Diagram

D) Prioritization Grid

Concrete Strength Failures



CORRECT ANSWER: C) Fishbone (Ishikawa) Diagram

WHY THIS TOOL?

The Cause-and-Effect (Fishbone/Ishikawa) Diagram is the definitive tool for root cause brainstorming. It structures potential causes into major branches (the 4Ms + Environment), prevents the team from jumping to solutions, and promotes deep 'five-why' questioning. No other B7 or N7 tool serves this specific purpose of visually mapping and grouping potential causes back to a central effect.



REAL-WORLD OUTCOME

The fishbone identified that incorrect water-cement ratio (Materials branch) combined with inadequate curing time (Method branch) were the co-root causes.

Choosing a Cloud Platform

SCENARIO

A fintech startup in Bahrain needs to select a cloud infrastructure provider. After initial research, the team shortlisted three providers: AWS, Microsoft Azure, and Google Cloud. The CTO identified five decision criteria: cost, security compliance, scalability, technical support quality, and regional data centre availability. Each criterion is not equally important — security and compliance is critical; cost is less so.

? The team needs to make a structured, multi-criteria decision by assigning different weights to each criterion and scoring all three alternatives consistently. Which N7 tool is designed for this?

OPTIONS

A) Affinity Diagram

B) Tree Diagram

C) Prioritization Grid

D) Activity Network Diagram

Choosing a Cloud Platform



CORRECT ANSWER: C) Prioritization Grid

WHY THIS TOOL?

A Prioritization Grid (Prioritization Matrix) is an N7 tool specifically designed for multi-criteria decision making with weighted criteria. It assigns weights to each criterion, scores each alternative against those criteria, and produces a ranked outcome. The affinity diagram organises ideas; the tree diagram breaks goals into tasks; the activity network diagram plans project schedules. Only the prioritization grid handles weighted multi-criteria ranking.



REAL-WORLD OUTCOME

Azure scored highest when security compliance was weighted at 40% — a result that would not have been obvious without the structured grid.

Improving a Visa Application Process

SCENARIO

A government immigration department wants to redesign its visa application process to reduce average processing time from 14 days to 5 days. A project team of 8 people from different sections (IT, legal, customer service, document verification) brainstorms improvements. After 20 minutes, the team has 60+ sticky notes with ideas, problems, and suggestions scattered across the table — and nobody agrees on how to organise them.

? The team needs to silently cluster 60+ unstructured ideas into natural theme groups before they can plan any solution. Which N7 tool is appropriate here?

OPTIONS

A) Process Decision Program Chart

B) Affinity Diagram

C) Pareto Chart

D) Interrelationship Digraph

Improving a Visa Application Process



CORRECT ANSWER: B) Affinity Diagram

WHY THIS TOOL?

An Affinity Diagram is the N7 tool designed to organise large volumes of unstructured qualitative data (ideas, opinions, issues) into natural theme groups through silent group consensus. Team members silently move cards until clusters emerge — producing organised themes that guide further analysis. The Pareto chart needs numerical frequency data; the PDPC plans against known risks; the digraph maps relationships between already-identified issues.



REAL-WORLD OUTCOME

The affinity diagram grouped 60 ideas into 7 themes — revealing that 'document duplication' was the most recurring theme, cutting 3 days off processing time.

Surgical Instrument Sterilisation Project

SCENARIO

A hospital is implementing a new centralised sterilisation unit for surgical instruments. The project involves 14 interdependent tasks across 4 departments — procurement, facilities, IT, and nursing training. Some tasks cannot start until others are finished. The project manager needs to ensure the unit is operational within 90 days and must identify which sequence of tasks, if delayed, would push back the entire project completion date.

? The project manager needs to map all tasks, their dependencies, durations, and identify the longest (critical) path to control the project schedule. Which quality/management tool is best suited?

OPTIONS

A) Tree Diagram

B) Matrix Diagram

C) Activity Network Diagram (PERT)

D) Process Decision Program Chart

Surgical Instrument Sterilisation Project



CORRECT ANSWER: C) Activity Network Diagram (PERT)

WHY THIS TOOL?

An Activity Network Diagram (PERT/CPM) is specifically designed for project scheduling with interdependent tasks. It maps tasks as nodes with durations and arrows showing dependencies, allowing calculation of the critical path — the longest sequence of dependent tasks that determines overall project duration. A tree diagram breaks goals into sub-tasks but doesn't show dependencies or time; a matrix shows relationships; a PDPC plans contingencies.



REAL-WORLD OUTCOME

The PERT diagram revealed the IT system integration task was on the critical path — the project manager assigned extra resources to it and the unit opened on day 88.

QUICK REFERENCE

B7 & N7 Tool Selection Guide

Don't know the process yet

Process Map

Root cause brainstorming

Fishbone Diagram

Need to count defect types

Check Sheet

Which problem to fix first?

Pareto Chart

Distribution of numeric data

Histogram

Organise 30+ ideas into themes

Affinity Diagram

Relationship between 2 variables

Scatter Diagram

Choose between options w/ criteria

Prioritization Grid

Process stable over time?

Control Chart

Project scheduling with dependencies

Activity Network (PERT)